Practical 03

Write a simple program in SCALA using Apache Spark framework

Before running this Scala program, ensure the following are installed:

- Java 8 or later (sudo apt install openjdk-11-jdk)
- Scala (sudo apt install scala)
- Apache Spark (download and extract from https://spark.apache.org/downloads.html)

Step 1: Create a Sample Input File

```
mkdir wordcount_input
echo "Apache Spark is fast. Spark is fun to use." > wordcount_input/input.txt
```

Step 2: Scala Program

```
import org.apache.spark.sql.SparkSession
object SimpleWordCount {
  def main(args: Array[String]): Unit = {
    // Create Spark session
    val spark = SparkSession.builder
      .appName("Simple Word Count")
      .master("local[*]") // run locally using all cores
      .getOrCreate()
    // Read input file
    val input = spark.sparkContext.textFile("wordcount_input/input.txt")

    // Perform Word Count
    val counts = input
      .flatMap(line => line.split(" "))
      .map(word => (word.toLowerCase().replaceAll("""[\p{Punct}]""", ""), 1))
      .reduceByKey(_ + _)
```

```
    // Save result
    counts.saveAsTextFile("wordcount_output")

    // Stop Spark
    spark.stop()
  }
}
```

Step 3: Compile and Run the Program

```
scalac -classpath $SPARK_HOME/jars/* SimpleWordCount.scala
```

```
$SPARK_HOME/bin/spark-submit \
  --class SimpleWordCount \
  --master local[*] \
  SimpleWordCount.jar
```

```
$SPARK_HOME/bin/spark-submit --class SimpleWordCount --master local[*] SimpleWordCount.scala
```

Step 4: Output

```
cat wordcount_output/part-*
```

```
(spark,2)
(is,2)
(fast,1)
(apache,1)
(fun,1)
(to,1)
(use,1)
```